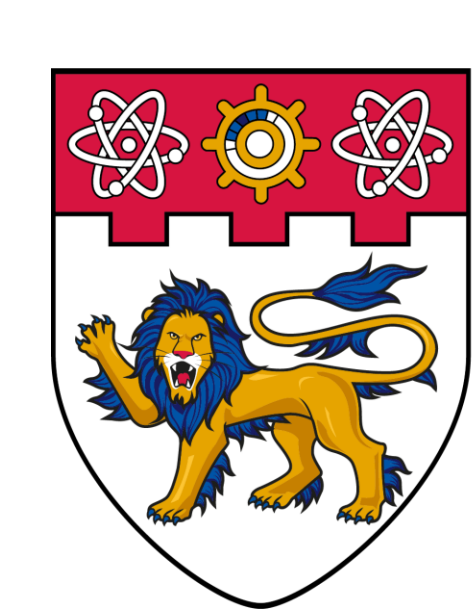
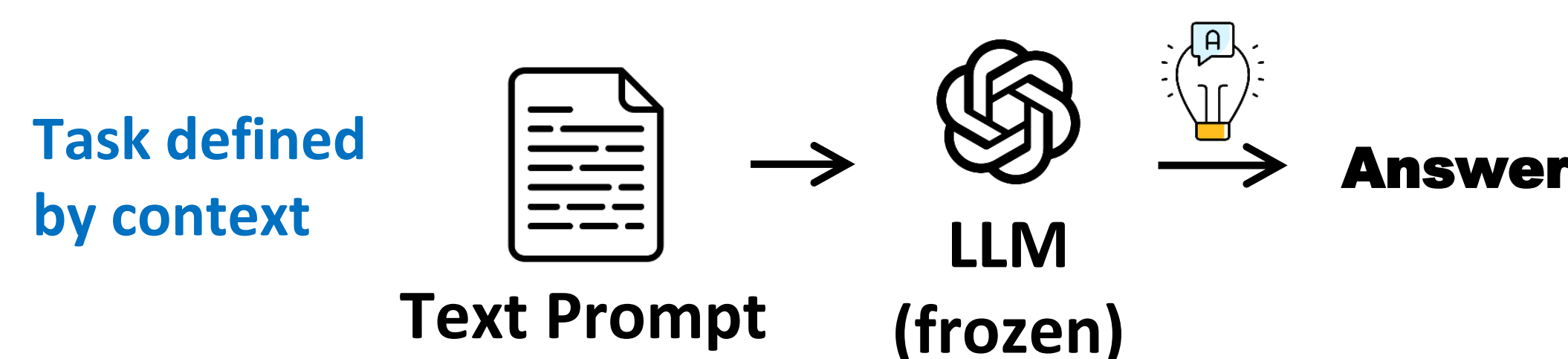


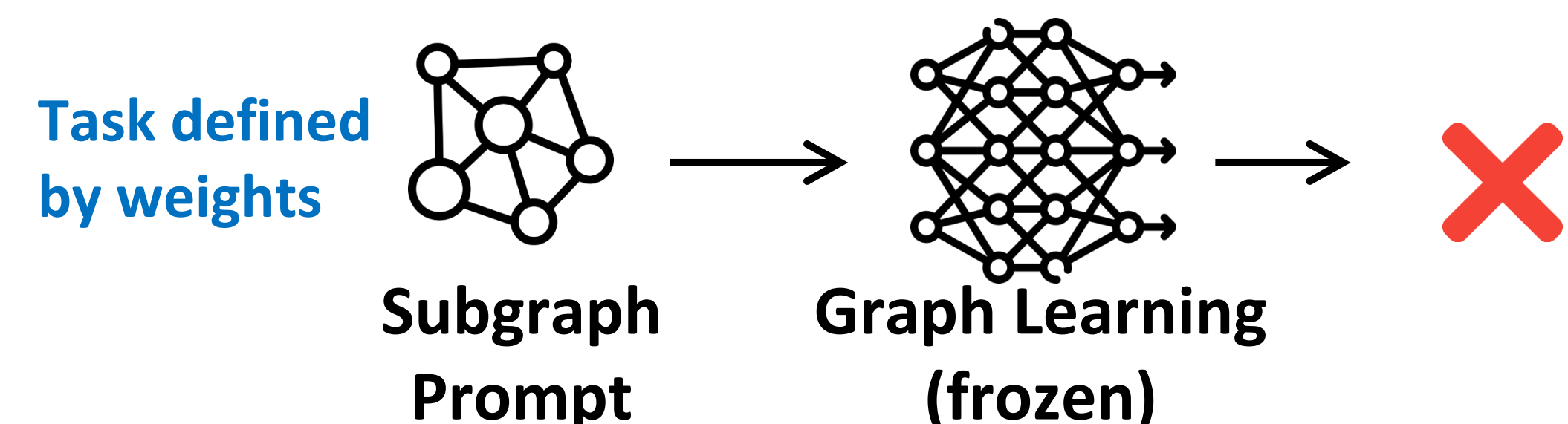


MOTIVATION

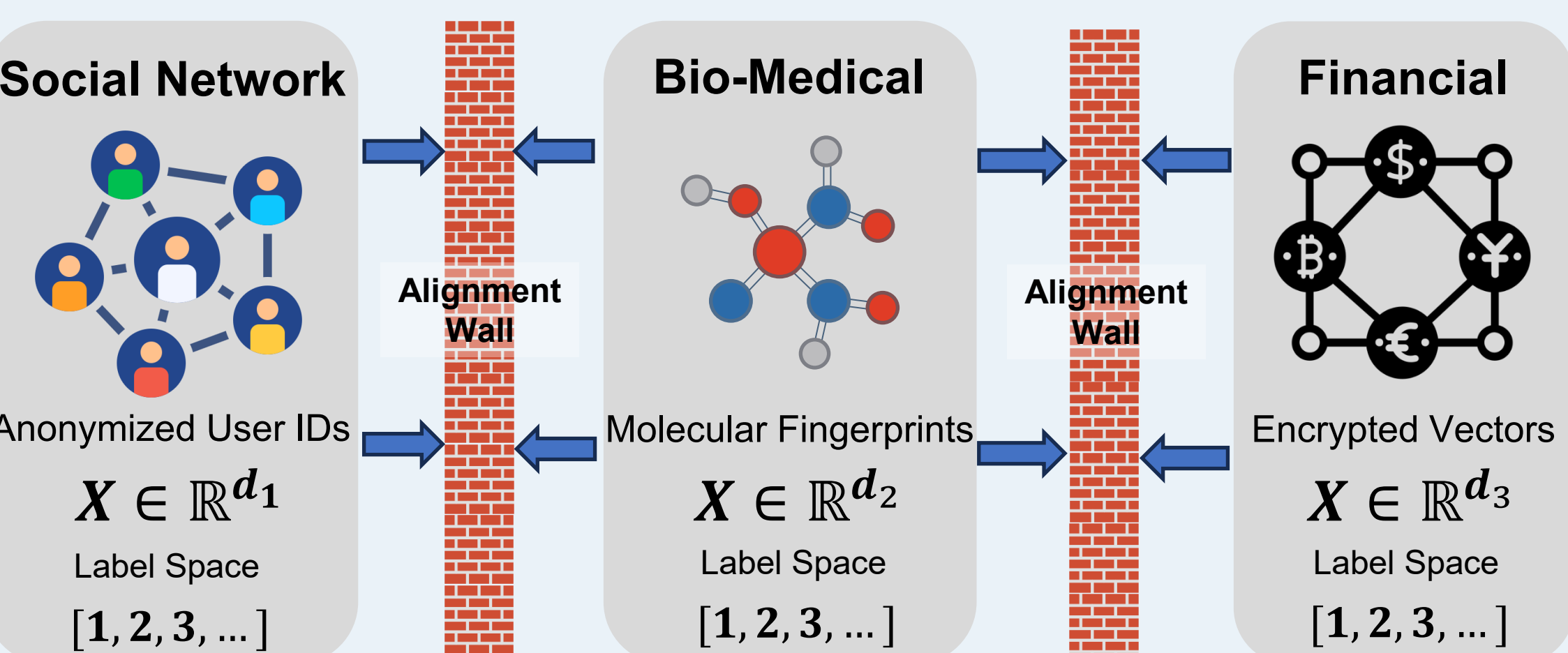
In-context learning (ICL) enables LLMs to adapt to new tasks from few examples without parameter updates. Text acts as a universal modality that naturally aligns diverse tasks and domains into a shared space.



But for graph learning models, tasks and domains are defined by trained weights — **graphs lack a unified modality like text for cross-domain alignment**, making true ICL on graphs elusive.



Impossible to reverse-engineer text from pre-encoded vectors

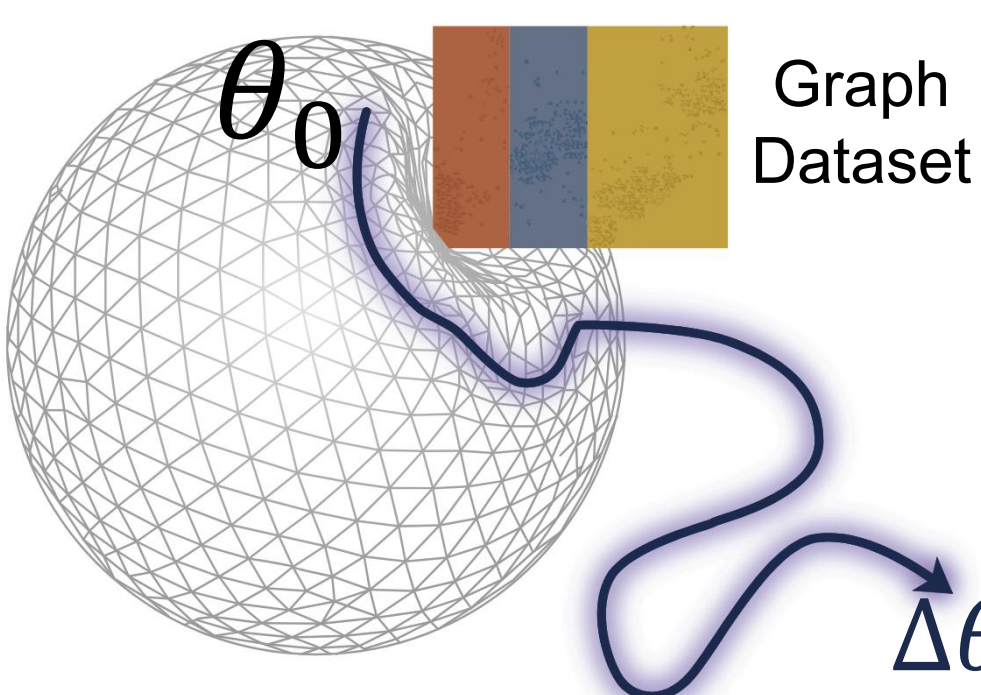


True In-Context Learning on Graphs

Method	Tuning-Free	Domain Alignment	Modality-Free
SSL-GNNs	✗	✗	✓
Prodigy	✓	✗	✓
UniGraph / OFA	✓	✓	✗
MF-GIA (Ours)	✓	✓	✓

Domain Embedder: Gradient Fingerprint

The gradient update captures how structure, features, and labels *jointly* influence the model, creating a unique identity for the domain.



Gradient Fingerprint:

$$\Delta\theta_i = \nabla L_i(\theta_0)$$

Domain Embedding:

$$e_i = \text{MLP}(\text{Conv2D}(\Delta\theta_i))$$

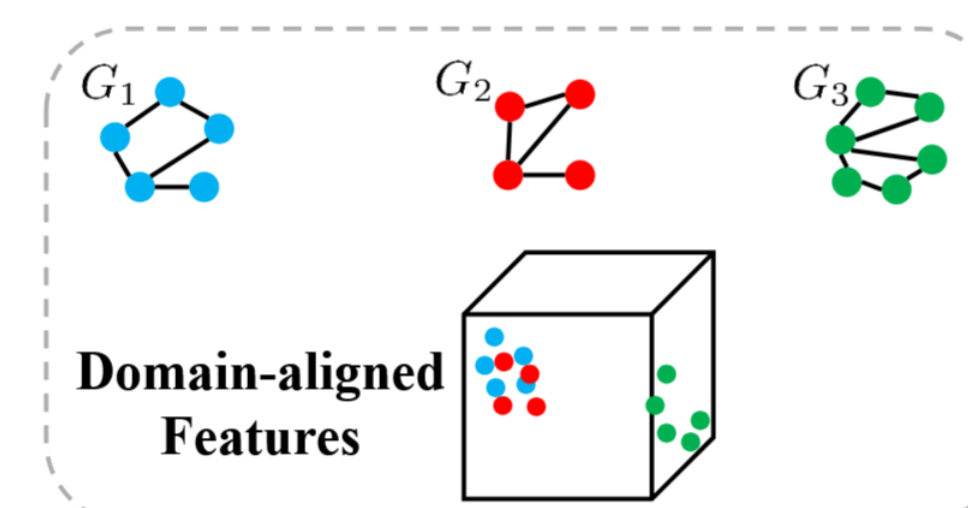
Modality-Free Alignment

Feature Alignment

The domain embedding modulates the feature space via FiLM:

$$z = \gamma(e_i) \odot x + \beta(e_i)$$

Nearby domains produce similar transforms → neighboring subspaces.

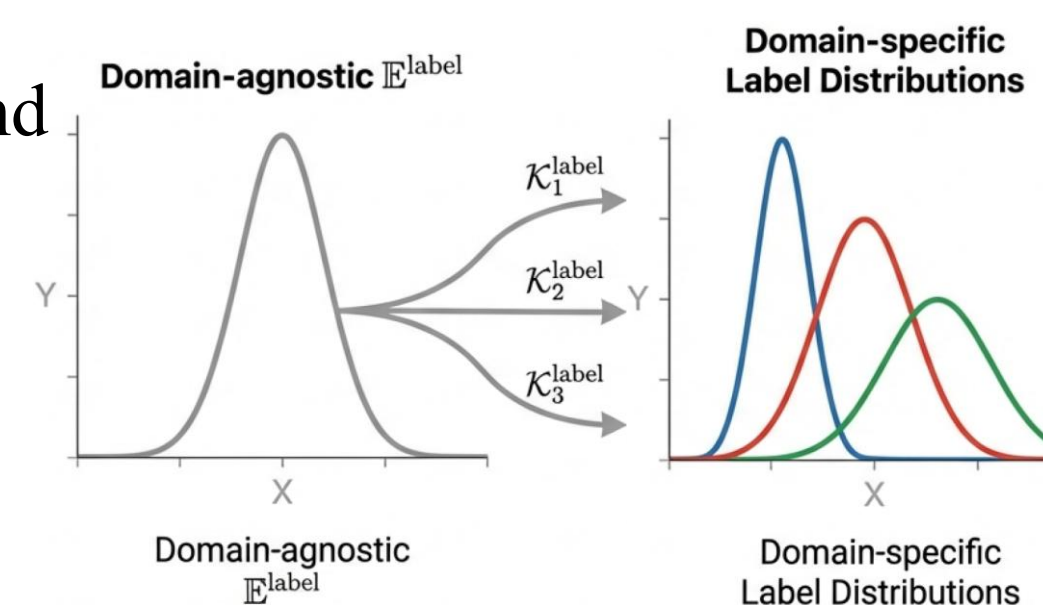


Label Alignment

Shared label prototypes are shifted and scaled conditioned on the domain embeddings:

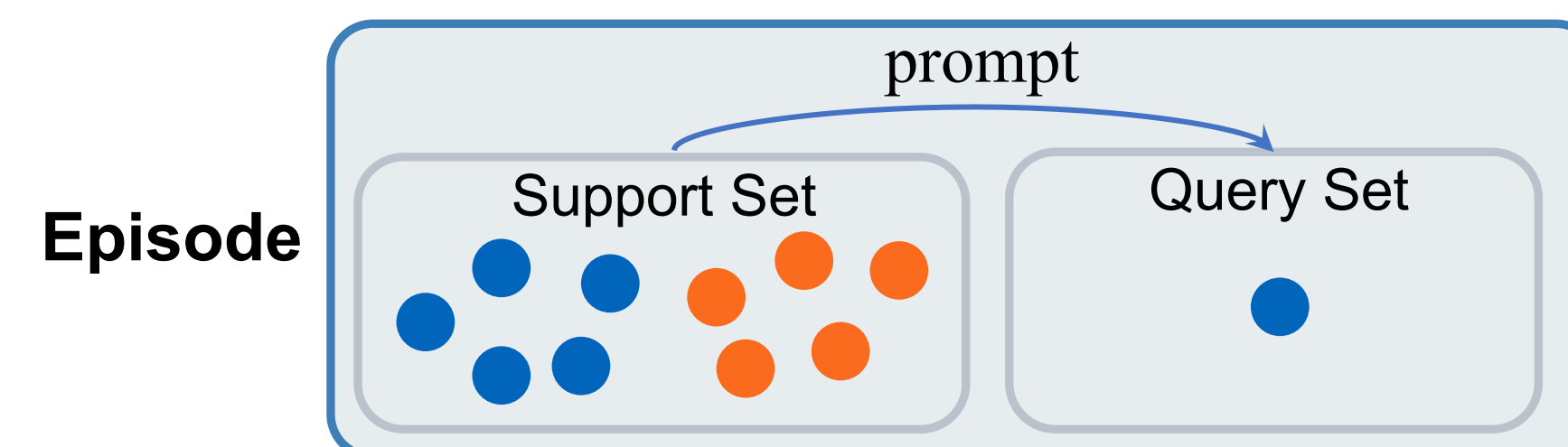
$$u = \gamma'(e_i) \odot E + \beta'(e_i)$$

Shared label base → domain-specific prototypes.



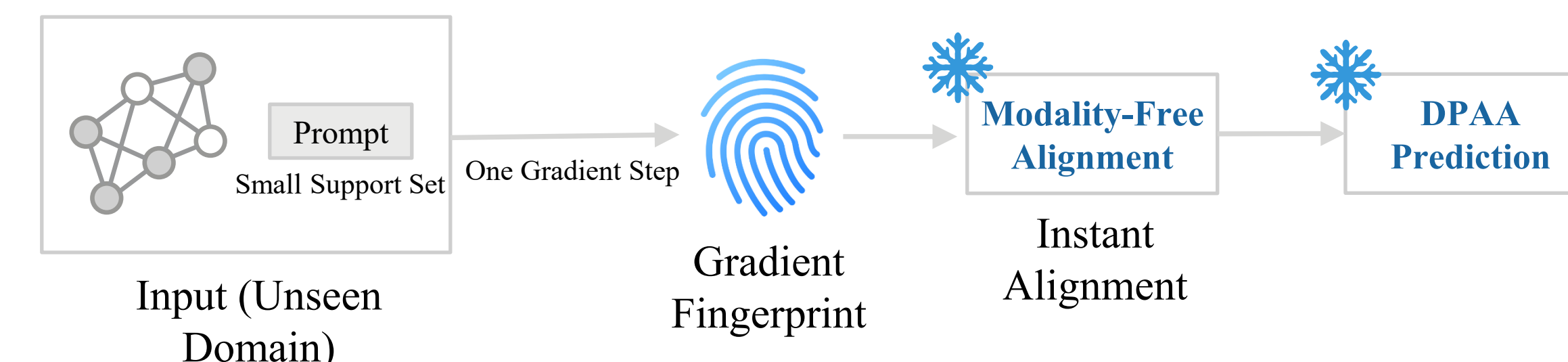
Episodic Pretraining

Dual Prompt-Aware Attention: Learning to Read Prompt



- Feature-side Attention: Query attends to support features to get the prompt-conditioned query representation.
- Label-side Attention: Prompt-conditioned prediction for the query.
- Prompts guide queries but do NOT interact with each other, strictly following ICL principles.

In-Context Prediction

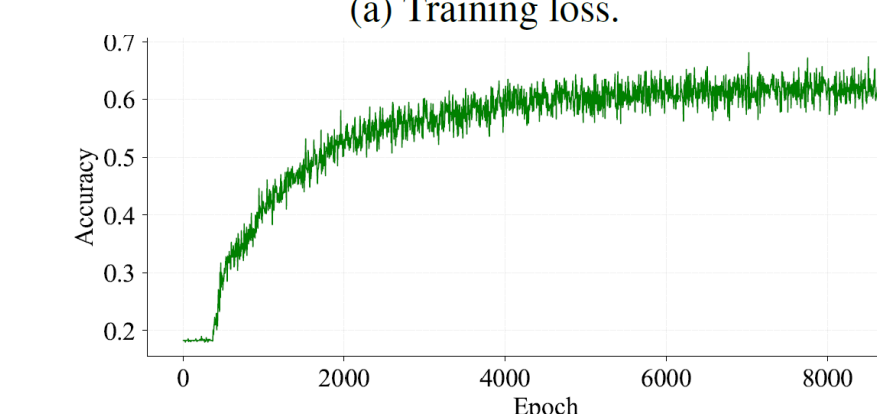
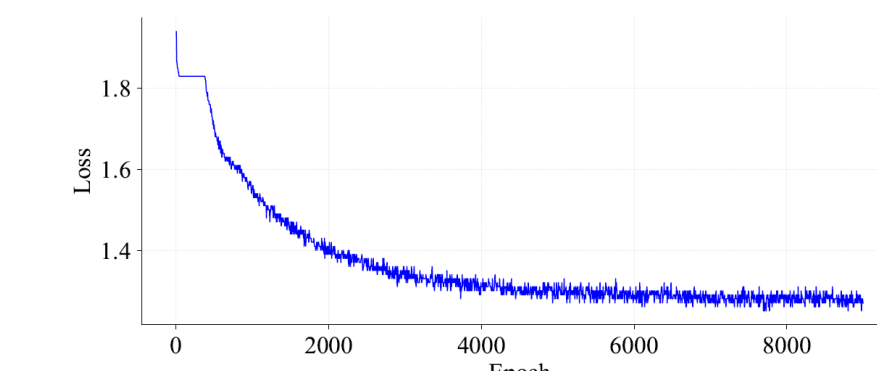


In-Context Prediction performs **parameter-update-free** inference on unseen domains.

Experiments

Datasets

Usage	Dataset	Domain
Pretrain	WikiCS	Web link
	PubMed	Citation
	ogbn-Arxiv	Citation
	Amazon-ratings	E-commerce (Ratings)
Evaluation	Cora	Citation
	ogbn-Products	E-commerce (Product Category)
	Computers	E-commerce (Product Category)
	Physics	Co-authorship
	BlogCatalog	Social Media
	FB15K237	Encyclopedic KG
	WN18RR	Lexical KG



Cross-Domain Generalization

Model	Products (E-Commerce)	Physics (Co-Author)	BlogCatalog (Social Media)
SAGE+FT	9.42	77.36	58.03
Prodigy	11.46	79.47	53.44
GFT	15.43	---	---
OFA	8.66	---	---
MF-GIA	22.61	88.92	67.31

Existing TAG-based models cannot run on pre-encoded graphs without raw text data.

Cross-Task Generation

Model	Accuracy
MAML	63.13%
Prodigy	71.45%
MF-GIA	91.38%

MF-GIA generalizes to entirely new structural reasoning tasks it never saw during pretraining.

